



PRODUCT NAME
FUROXIME INJECTION 750 MG

NAME AND STRENGTH OF ACTIVE SUBSTANCE
Each vial contains: 750 mg cefuroxime as cefuroxime sodium

PRODUCT DESCRIPTION
White sterile powder in clear glass vials

PHARMACODYNAMICS
Pharmacotherapeutic group: antibacterials for systemic use, second-generation cephalosporins

Mechanism of Action
Cefuroxime inhibits bacterial cell wall synthesis following attachment to penicillin binding proteins. This results in the interruption of cell wall (peptidoglycan) biosynthesis, which leads to bacterial cell lysis and death.

Microbiological Susceptibility
The prevalence of acquired resistance may vary geographically and time dependent for selected species, local information on resistance is desirable, particularly when treating severe infections.

Cefuroxime is usually active against the following microorganisms in vitro (*asterisk indicates clinical efficacy of cefuroxime has been demonstrated in clinical trials):-

COMMONLY SUSCEPTIBLE SPECIES	
Gram-Positive Aerobes: <i>Streptococcus pyogenes</i> Beta-hemolytic streptococci	Gram-Negative Aerobes: <i>Haemophilus influenzae</i> (including ampicillin resistant strains) <i>Haemophilus parainfluenzae</i> <i>Moraxella catarrhalis</i> <i>Neisseria gonorrhoea</i> (including penicillinase and non-penicillinase producing strains) <i>Neisseria meningitidis</i> <i>Shigella</i> spp.
Gram-Positive Anaerobes: <i>Peptostreptococcus</i> spp. <i>Propionibacterium</i> spp.	Spirochetes: <i>Borrelia burgdorferi</i>

ORGANISMS FOR WHICH ACQUIRED RESISTANCE MAY BE A PROBLEM	
Gram-Positive Aerobes: <i>Staphylococcus</i> spp. including <i>S. aureus</i> (methicillin-susceptible isolates only) <i>Streptococcus pneumoniae</i> Viridans group streptococcus	Gram-Negative Aerobes: <i>Bordetella pertussis</i> <i>Citrobacter</i> spp. not including <i>C. freundii</i> <i>Enterobacter</i> spp. not including <i>E. aerogenes</i> and <i>E. cloacae</i> <i>Escherichia coli</i> <i>Klebsiella</i> spp. including <i>K. pneumoniae</i> <i>Proteus mirabilis</i> <i>Proteus</i> spp. not including <i>P. penneri</i> and <i>P. vulgaris</i> <i>Providencia</i> spp. <i>Salmonella</i> spp.
Gram-Positive Anaerobes: <i>Clostridium</i> spp. not including <i>C. difficile</i>	Gram-Negative Anaerobes: <i>Bacteroides</i> spp. not including <i>B. fragilis</i> <i>Fusobacterium</i> spp.

INHERENTLY RESISTANT ORGANISMS	
Gram-Positive Aerobes: <i>Enterococcus</i> spp. including <i>E. faecalis</i> and <i>E. faecium</i> <i>Listeria monocytogenes</i>	Gram-Negative Aerobes: <i>Acinetobacter</i> spp. <i>Burkholderia cepacia</i> <i>Campylobacter</i> spp. <i>Citrobacter freundii</i> <i>Enterobacter aerogenes</i> <i>Enterobacter cloacae</i> <i>Morganella morganii</i> <i>Proteus penneri</i> <i>Proteus vulgaris</i> <i>Pseudomonas</i> spp. including <i>P. aeruginosa</i> <i>Serratia</i> spp. <i>Stenotrophomonas maltophilia</i>
Gram-Positive Anaerobes: <i>Clostridium difficile</i>	Gram-Negative Anaerobes: <i>Bacteroides fragilis</i>
Others: <i>Chlamydia</i> spp. <i>Mycoplasma</i> spp. <i>Legionella</i> spp.	

PHARMACOKINETICS

Absorption
After intramuscular (IM) injection of cefuroxime 750 and 1000 mg doses to normal volunteers, the mean peak was achieved within 30 to 60 minutes. For intravenous (IV) doses of 750 and 1500 mg, serum concentrations were approximately 50 and 100 µg/mL respectively, at 15 minutes. There was no evidence of accumulation of cefuroxime in the serum following repeat IV administration of 1500 mg doses every 8 hours in normal volunteers.

Distribution
Protein binding has been stated as 33 to 50%, depending on the methodology used. The average volume of distribution ranges from 9.3 to 15.8 L/1.73 m² following IM or IV administration over the dosage range of 250 to 1000 mg. Cefuroxime passes the blood-brain barrier when the meninges are inflamed.

Elimination
Cefuroxime is excreted by glomerular filtration and tubular secretion. The serum half-life is approximately 70 minutes following IV or IM administration. There is an almost complete recovery (85 to 90%) of unchanged cefuroxime in urine within 24 hours of administration. Majority of the cefuroxime is excreted within the first 6 hours.

Special Patient Populations

Elderly
Following IM or IV administration, pharmacokinetics profiles (absorption, distribution, excretion) of cefuroxime in elderly patients are similar to younger patients with equivalent renal function.

Paediatrics
The serum half-life of cefuroxime has been shown to be substantially prolonged in neonates according to gestational age. However, in older infants (aged > 3 weeks) and in children, the serum half-life is similar to that observed in adults (60 to 90 minutes).

- INDICATION**
- Respiratory tract infections (e.g. acute/chronic bronchitis, infected bronchiectasis, bacterial pneumonia, lung abscess and post-operative chest infections)
 - Ear, nose and throat infections (e.g. sinusitis, tonsillitis, pharyngitis and otitis media)
 - Urinary tract infections (e.g. acute/chronic pyelonephritis, cystitis and asymptomatic bacteriuria)
 - Soft tissue infections (e.g. cellulitis, erysipelas and wound infections)
 - Bone and joint infections (e.g. osteomyelitis and septic arthritis)
 - Obstetric and gynaecological infections, pelvic inflammatory diseases
 - Gonorrhea particularly when penicillin is unsuitable
 - Other infections including septicaemia, meningitis and peritonitis
 - Prophylaxis against infection in abdominal, pelvic, orthopaedic, cardiac, pulmonary, oesophageal and vascular surgery where there is increased risk from infection

- RECOMMENDED DOSAGE**
- General Dosage Recommendations**
- Adults**
- Many infections will respond to 750 mg every 8 hours by intramuscular or intravenous injection
 - For more severe infections, the dose should be increased to 1.5 g every 8 hours intravenously; the frequency of IV or IM injections can be increased to 6 hourly if necessary, giving total doses of 3 to 6 g daily
- Infants and Children**
- 30 to 100 mg/kg/day given as 3 or 4 divided doses
 - A dose of 60 mg/kg/day will be appropriate for most infections
- Neonates**
- 30 to 100 mg/kg/day given as 2 or 3 divided doses (See *Pharmacokinetics*)
- Gonorrhoea**
- Adults**
- 1.5 g as a single dose (as 2 x 750 mg injections given intramuscularly with different sites)
- Meningitis**
- Adults**
- 3 g given intravenously every 8 hours
- Infants and Children**
- 150 to 250 mg/kg/day given intravenously in 3 or 4 divided doses
- Neonates**
- 100 mg/kg/day given intravenously
- Prophylaxis**
- The usual dose is 1.5 g given intravenously with induction of anaesthesia for abdominal, pelvic and orthopedic operations and may be supplemented with 2 x 750 mg doses given intramuscularly at 8 and 16 hours later
 - In cardiac, pulmonary, oesophageal and vascular operations, the usual dose is 1.5 g given intravenously with induction of anaesthesia, continuing with 750 mg given intramuscularly every 8 hours for a further 24 to 48 hours
 - In total joint replacement, 1.5 g cefuroxime may be mixed dry with each pack of methyl methacrylate cement polymer before adding the liquid monomer

Renal Impairment
Cefuroxime is primarily excreted by the kidneys, therefore, dose adjustment is recommended in patients with markedly impaired renal function to compensate for its slower excretion.

Creatinine Clearance	Recommended Dose
> 20 mL/min	It is not necessary to reduce the standard dose (750 mg to 1.5 g every 8 hours)
10 to 20 mL/min	750 mg every 12 hours
< 10 mL/min	750 mg every 24 hours
Patients on haemodialysis	- A further 750 mg dose should be given intravenously or intramuscularly at the end of each dialysis - In addition to parenteral use, cefuroxime can be incorporated into the peritoneal dialysis fluid (usually 250 mg for every 2 litres of dialysis fluid)
Patients in renal failure, on continuous arteriovenous haemodialysis or high-flux haemofiltration in intensive therapy units	750 mg every 12 hours, for low-flux haemofiltration follow the dosage recommended under impaired renal function

ROUTE OF ADMINISTRATION

For intravenous (IV) and/or intramuscular (IM) administration

Intravenous
Dissolve Furoxime in Water for Injections using at least 6 mL for 750 mg, or 15 mL for 1.5 g.

Intravenous Infusion
Dissolve 1.5 g of Furoxime in 15 mL Water for Injections. Add the reconstituted solution to 50 to 100 mL of a compatible infusion fluid (See *Compatibilities*). These solutions may be given directly into the vein or introduced into the tubing of the giving set if the patient is receiving parenteral fluids.

Intramuscular
Add 3 mL sterile water for injections to 750 mg Furoxime. Shake gently to produce an opaque suspension.

CONTRAINDICATIONS

Hypersensitivity to cephalosporin antibiotics

WARNINGS AND PRECAUTIONS

Hypersensitivity Reactions

As with all beta-lactam antibacterial agents, serious and occasionally fatal hypersensitivity reactions have been reported. In case of severe hypersensitivity reactions, treatment with cefuroxime must be discontinued immediately and adequate emergency measures must be initiated.

Before beginning treatment, it should be established whether the patient has a history of severe hypersensitivity reactions to cefuroxime, to other cephalosporins or to any other type of beta-lactam agent. Caution should be used if cefuroxime is given to patients with a history of non-severe hypersensitivity to other beta-lactam agents.

Concurrent Treatment with Potent Diuretics or Aminoglycosides
Cephalosporin antibiotics at high dosage should be given with caution to patients receiving concurrent treatment with potent diuretics such as furosemide or aminoglycosides. Renal impairment has been reported during use of these combinations. Renal function should be monitored in the elderly and those with known pre-existing renal impairment (See *Recommended Dosage*).

Overgrowth of Non-Susceptible Microorganisms
Use of cefuroxime may result in the overgrowth of *Candida*. Prolonged use may also result in the overgrowth of other non-susceptible microorganisms (e.g. *enterococci* and *Clostridium difficile*). Antibacterial agent-associated pseudomembranous colitis has been reported with use of cefuroxime and may range in severity from mild to life threatening. This diagnosis should be considered in patients with diarrhoea during or subsequent to the administration of cefuroxime. Discontinuation of therapy with cefuroxime and the administration of specific treatment for *Clostridium difficile* should be considered. Medicinal products that inhibit peristalsis should not be given.

EFFECTS ON THE ABILITY TO DRIVE AND USE MACHINES

None reported. Cefuroxime is unlikely to have an effect on the ability to drive and use machines.

INTERACTIONS WITH OTHER MEDICAMENTS

In common with other antibiotics, cefuroxime may affect the gut flora, leading to lower oestrogen reabsorption and reduced efficacy of combined oral contraceptives.

Cefuroxime is excreted by glomerular filtration and tubular secretion. Concomitant use of probenecid is not recommended. Concurrent administration of probenecid prolongs the excretion of the antibiotic and produces an elevated peak serum level.

High-dosage treatments with cephalosporins should be carried out with caution on patients who are taking strong-acting diuretics (such as furosemide) or potential nephrotoxic preparations (such as aminoglycoside antibiotics), since impairment of renal function through such combinations cannot be ruled out. Concomitant use with oral anticoagulants may give rise to increased international normalized ratio (INR).

PREGNANCY AND LACTATION

Pregnancy

There are limited amounts of data from the use of cefuroxime in pregnant women. Studies in animals have shown no reproductive toxicity. As with all drugs, cefuroxime should be administered with caution during the early months of pregnancy and only if the benefit outweighs the risk.

Lactation

Cefuroxime is excreted in human milk in small quantities, and consequently caution should be exercised when the medicinal product is administered in a nursing mother.

UNDESIRABLE EFFECTS

Summary of Safety Profile

The most common adverse reactions are neutropenia, eosinophilia, transient rise in liver enzymes or bilirubin, particularly in patients with pre-existing liver disease, but there is no evidence of harm to the liver and injection site reactions. The adverse reactions presented below are classified by system organ and frequency in decreasing order of the seriousness.

System Organ	Frequency	Adverse Reactions
Infections and infestations	Not known	: Candida overgrowth, overgrowth of <i>Clostridium difficile</i>
Blood and lymphatic system disorders	Common	: Neutropenia, eosinophilia, decreased haemoglobin concentration
	Uncommon	: Leukopenia, positive Coomb's test
	Not known	: Thrombocytopenia, haemolytic anaemia
Immune system disorders	Not known	: Drug fever, interstitial nephritis, anaphylaxis, cutaneous vasculitis
Gastrointestinal disorders	Uncommon	: Gastrointestinal disturbance
	Not known	: Pseudomembranous colitis
Hepatobiliary disorders	Common	: Transient rise in liver enzymes
	Uncommon	: Transient rise in bilirubin
Skin and subcutaneous tissue disorders	Uncommon	: Skin rash, urticaria and pruritus
	Not known	: Erythema multiforme, toxic epidermal necrolysis and Stevens-Johnson syndrome, angioneurotic oedema

Renal and urinary disorders	Not known	: Elevations in serum creatinine, elevations in blood urea nitrogen and decreased creatinine clearance
General disorders and administration site conditions	Common	: Injection site reactions which may include pain and thrombophlebitis

Description of selected adverse reactions

- Cephalosporins as a class tend to be absorbed onto the surface of red cell membranes and react with antibodies directed against the drug to produce a positive Coomb's test (which can interfere with cross matching of blood) and very rarely haemolytic anaemia
- Transient rises in serum liver enzymes or bilirubin have been observed which are usually reversible
- Pain at the intramuscular injection site is more likely at higher doses. However it is unlikely to be a cause for discontinuation of treatment

Paediatric Population

The safety profile for cefuroxime sodium in children is consistent with the profile in adults.

OVERDOSE AND TREATMENT

Overdose can lead to neurological sequelae including encephalopathy, convulsions and coma. Symptoms of overdose can occur if the dose is not reduced appropriately in patients with renal impairment (See *Recommended Dosage*).

Serum levels of cefuroxime can be reduced by haemodialysis or peritoneal dialysis.

INCOMPATIBILITIES

Furoxime should not be mixed in the syringe with aminoglycoside antibiotics. The pH of 2.74% W/V Sodium Bicarbonate Injection BP considerably affects the colour of the solutions and therefore this solution is not recommended for the dilution of Furoxime. However, if required, for patients receiving Sodium Bicarbonate Injection by infusion, Furoxime may be introduced into the tube of the giving set.

COMPATIBILITIES

Concentration (mg/mL)	Solvent / Vial	Time / Condition	
		25°C	2-8°C
220	Sterile Water for Injection 3 mL	8 hrs	48 hrs
90	Sterile Water for Injection 8 mL	8 hrs	48 hrs
30	Sterile Water for Injection 25 mL	7 hrs	7 days
30	0.9% Sodium Chloride Injection 25 mL	7 hrs	7 days
30	5% Dextrose Injection 25 mL	7 hrs	7 days
30	Lactate Ringer's Injection 25 mL	7 hrs	7 days

STORAGE CONDITIONS

Store below 30°C. Protect from light.

The storage conditions are detailed on the packaging.

Store in original package to protect from light.

Some increase in the colour of prepared solutions and suspensions of Furoxime may occur on storage.

SHELF LIFE

2 years. The expiry date of the powder is indicated on the packaging.

The unopened pack storage temperature requirements are indicated on the packaging.

If unopened product previously stored below 30°C

Shelf life after reconstitution and dilution under controlled and validated aseptic conditions:

Suspension and solution for Injection - 8 hours below 25°C or 72 hours at 2° to 8°C.

Solution for Infusion - Use immediately or within 24 hours at 2° to 8°C.

Shelf life if reconstitution and dilution has not taken place in controlled and validated aseptic conditions:

The product should be used immediately or within 24 hours if stored at 2° to 8°C.

DOSAGE FORMS AND PACKAGING AVAILABLE

Box of: 750 mg x 1 vial

DATE OF REVISION OF PI

Date of revision: 28/08/2023

FURO-750-03-L-ENG-MY-A0003

Copyright © 2023 Siam Bheasach. All rights reserved.

Manufactured by:- SIAM BHEASACH CO., LTD.
123 Soi Chokechaiummitr, Vibhavadi-Rangsit Road, Chomphon, Chatuchak, Bangkok 10900 and 9 Soi Chokechaiummitr 3, Vibhavadi-Rangsit Road, Dindang, Dindang, Bangkok 10400, Thailand.

Product Registration Holder, Importer:- SPG PHARMA (MALAYSIA) SDN. BHD.
Suite 1701, Block A, Damansara Intan e-Business Park,
No. 1, Jalan SS20/27, 47400 Petaling Jaya, Selangor Darul Ehsan, Malaysia.